

Experiment 2 :Netflix HLD

Requirements

Estimations

Non-Functional

- System should support 100 Million users
- The system must maintain 99.99% availability
- User data should be stored securely

Functional

- User can create account and login
- User can browse movies and shows
- User can search movies/shows
- User can watchlist a movie

Assumptions:

Time Period = 5 years
Registered Users = 100 Million
Daily Active Users = 10 Million
Sessions/User/Day = 2
Requests/Session = 20
Read : Write = 100 : 1
Peak Traffic ≈ 14,000 QPS
Read QPS ≈ 13,860/sec
Write QPS ≈ 140/sec

Estimations :

Cache = 64 GB
User Storage ≈ 150 GB
Watch History ≈ 37 TB
Movie Metadata ≈ 1 GB
Video Storage ≈ 250 TB
Total Storage ≈ 300 TB
Application Servers = 9
Database Servers = 4 (1 Write + 3 Read)

DataBase

User Table

User_ID	Name	Email	Password
INT (PRIMARY KEY)	VARCHAR	VARCHAR	VARCHAR

Watchlist Table

Watchlist_ID	User_ID	Movie_ID
INT (PRIMARY KEY)	INT	INT

Movie Table

Movie_ID	Title	Genre	Language	Release_Year
INT (PRIMARY KEY)	VARCHAR	VARCHAR	VARCHAR	INT

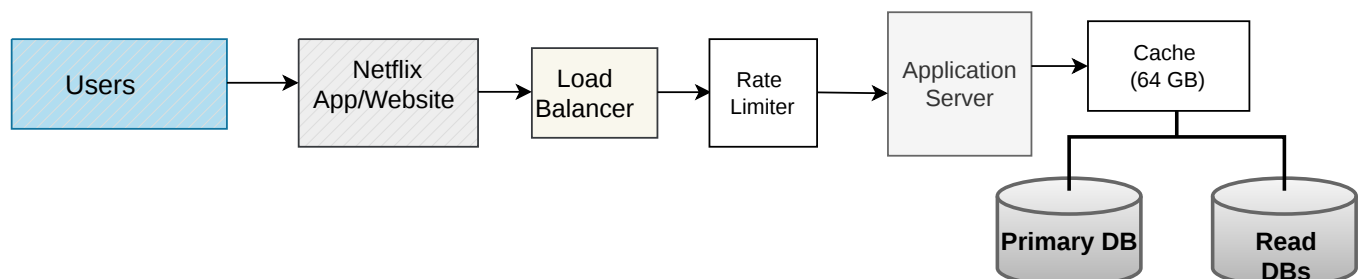
WatchHistory Table

history_id	User_ID	Movie_ID	Watch_Time
INT (PRIMARY KEY)	INT	INT	DATETIME

Relationships:

- **User** → **Watchlist**: One user can have many watchlist entries.
- **Movie** → **Watchlist**: One movie can be added to many users' watchlists.
- **User** → **Watch History**: One user can have many history entries.
- **Movie** → **Watch History**: One movie can appear in many users' watch histories.

Architecture



API Endpoints

Method	Endpoint	Purpose	Table
POST	/api/users	Register a new user	User
POST	/api/users/login	Login user	User
GET	/api/users/{userId}	Get user details	User
GET	/api/movies	Get all movies	Movie
GET	/api/movies/{movieId}	Get movie details	Movie
GET	/api/movies/search?title=abc	Search movies	Movie
POST	/api/watchlist	Add movie to watchlist	Watchlist
GET	/api/users/{userId}/watchlist	Get user's watchlist	Watchlist
DELETE	/api/watchlist/{watchlistId}	Remove movie from watchlist	Watchlist
POST	/api/history	Add movie to watch history	Watch History
GET	/api/users/{userId}/history	Get user's watch history	Watch History

API Flow:

